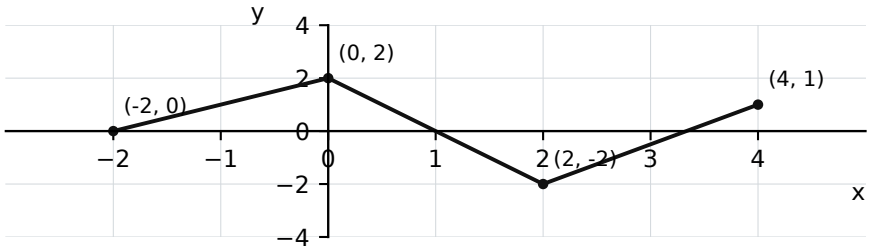
MATH 101 **34260**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 1);\ (3,\ 5)$	1. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
2. For this function, find f(-1), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 2\|x - 1\| - 2$	2. (a) 6 pts (b) 7 pts (c) 7 pts
3. From the graph, find average rate from x = -2 to x = 4; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	3. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts

Test continues on the other side of the sheet.

MATH 101 **34260**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (2x+1)^2 + 3$

A. $f(x) = 2x+1,\ g(x) = x^2+3$

B. $f(x) = 2x^2+3,\ g(x) = x+1$

C. $f(x) = x^2+1,\ g(x) = 2x+3$

D. $f(x) = x^2+3,\ g(x) = 2x+1$

(*b*) $F(x) = x^2,\ G(x) = x-2,\ x_0 = 3$

4. (a)

8 pts

(b)

8 pts

(c)

4 pts

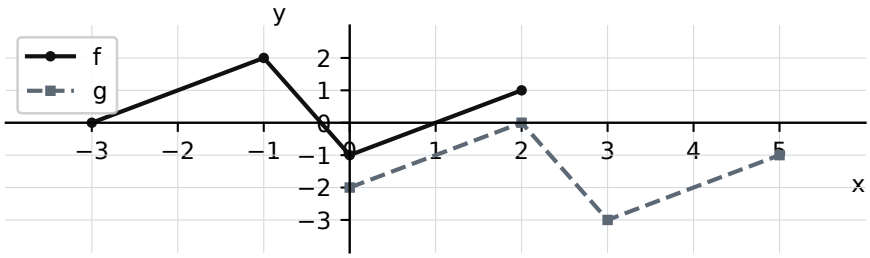
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-3) - 2$

B. $g(x) = f(x+3) + 2$

C. $g(x) = f(x+3) - 2$

D. $g(x) = f(x-3) + 2$



5.

20 pts

34260

TEST PAGE 1

1. (a)

$f(x) = 2x - 1$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{2} + \frac{1}{2}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

2. (a)

2 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-2, \infty)$ — 7 pts

3. (a)

$\frac{1}{6}$ — 6 pts

(b)

$(-2, 0) \cup (2, 4)$ — 4 pts

(c)

$(0, 2)$ — 4 pts

(d)

$(0,\ 2)$ — 3 pts

(e)

$(2,\ -2)$ — 3 pts

TEST PAGE 2

4. (a)

D

8 pts

(b)

$x^2 - 4x + 4$

8 pts

(c)

1

4 pts

5.

A

20 pts

MATH 101 **37133**

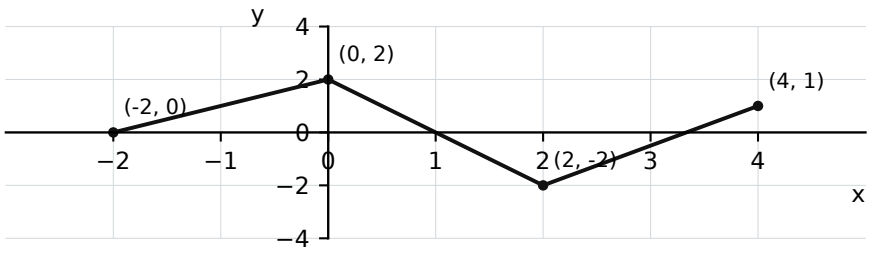
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. From the graph, find average rate from x = -2 to x = 4; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



1. (a) 6 pts

(b) 4 pts

(c) 4 pts

(d) 3 pts

(e) 3 pts

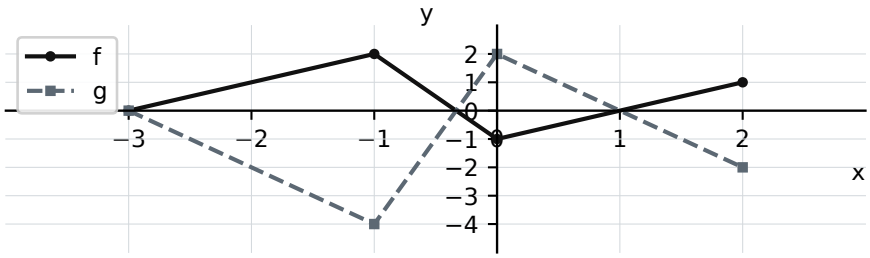
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = -2f(x)$

$B.\quad g(x) = f(x) - 2$

$C.\quad g(x) = f(-2x)$

$D.\quad g(x) = 2f(x)$



2. 20 pts

3. For this function, find f(-1), domain, and range. Record: (a) value; (b) domain; (c) range.

$f(x) = 2|x - 1| - 2$

3. (a) 6 pts

(b) 7 pts

(c) 7 pts

Test continues on the other side of the sheet.

MATH 101 **37133**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (2x+1)^2 + 3$ $A.\quad f(x) = 2x+1,\ g(x) = x^2+3$ $B.\quad f(x) = 2x^2+3,\ g(x) = x+1$ $C.\quad f(x) = x^2+1,\ g(x) = 2x+3$ $D.\quad f(x) = x^2+3,\ g(x) = 2x+1$ $(b)\quad F(x) = x^2,\ G(x) = x-2,\ x_0 = 3$	4. (a) 8 pts (b) 8 pts (c) 4 pts
5. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 1);\ (3,\ 5)$	5. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts

37133

TEST PAGE 1

1. (a)

$\frac{1}{6}$ 6 pts

(b)

$(-2, 0) \cup (2, 4)$ 4 pts

(c)

$(0, 2)$ 4 pts

(d)

$(0,\ 2)$ 3 pts

(e)

$(2,\ -2)$ 3 pts

2.

A

20 pts

3. (a)

2 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-2, \infty)$ 7 pts

TEST PAGE 2

4. (a)

D

8 pts

(b)

$x^2 - 4x + 4$

8 pts

(c)

1

4 pts

5. (a)

$f(x) = 2x - 1$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{2} + \frac{1}{2}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

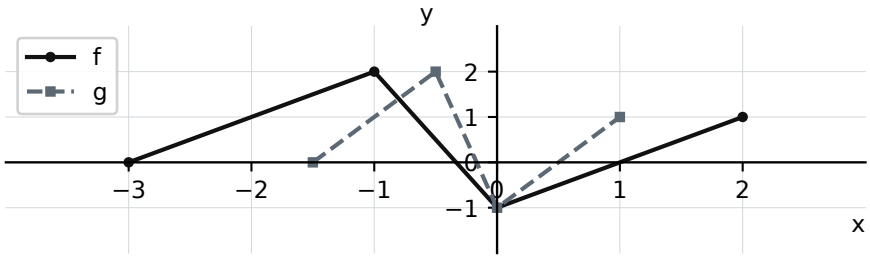
MATH 101 **94721**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 1);\ (3,\ 5)$	1. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
2. For this function, find f(-1), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 2\|x - 1\| - 2$	2. (a) 6 pts (b) 7 pts (c) 7 pts
3. The solid graph is f and the dashed graph is g. Select the expression for g(x). *A*. $g(x) = f(2x)$ *B*. $g(x) = f(x - 2)$ *C*. $g(x) = 2f(x)$ *D*. $g(x) = f(x/2)$ 	3. 20 pts

Test continues on the other side of the sheet.

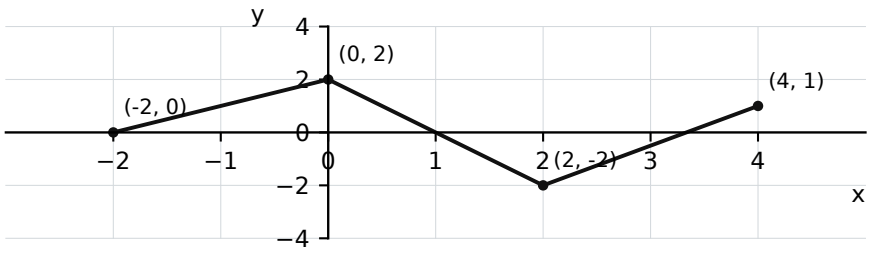
MATH 101 **94721**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -2 to x = 4; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	4. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts
5. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (2x+1)^2 + 3$ $A.\quad f(x) = 2x+1,\ g(x) = x^2+3$ $B.\quad f(x) = 2x^2+3,\ g(x) = x+1$ $C.\quad f(x) = x^2+1,\ g(x) = 2x+3$ $D.\quad f(x) = x^2+3,\ g(x) = 2x+1$ $(b)\quad F(x) = x^2,\ G(x) = x-2,\ x_0 = 3$	5. (a) 8 pts (b) 8 pts (c) 4 pts

94721

TEST PAGE 1

1. (a)

$f(x) = 2x - 1$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{2} + \frac{1}{2}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

2. (a)

2 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-2, \infty)$ 7 pts

3.

A

20 pts

TEST PAGE 2

4. (a)

$\frac{1}{6}$ 6 pts

(b)

$(-2, 0) \cup (2, 4)$ 4 pts

(c)

$(0, 2)$ 4 pts

(d)

$(0,\ 2)$ 3 pts

(e)

$(2,\ -2)$ 3 pts

5. (a)

D

8 pts

(b)

$x^2 - 4x + 4$

8 pts

(c)

1

4 pts

MATH 101 **25229**

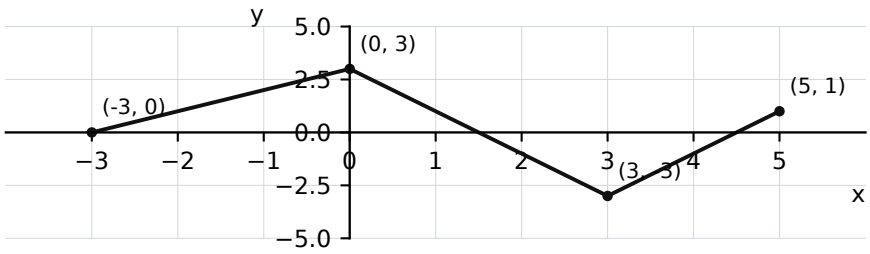
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. From the graph, find average rate from x = -3 to x = 5; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



1. (a) 6 pts

(b) 4 pts

(c) 4 pts

(d) 3 pts

(e) 3 pts

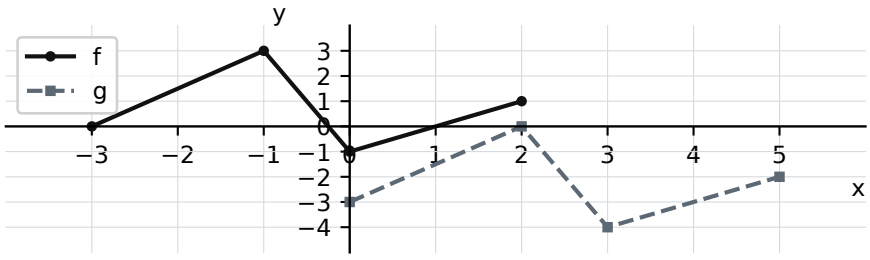
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x+3)+3$

$B.\quad g(x) = f(x-3)-3$

$C.\quad g(x) = f(x-3)+3$

$D.\quad g(x) = f(x+3)-3$



2. 20 pts

3. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 3);\ (3,\ 9)$

3. (a) 10 pts

(b) 6 pts

(c) 2 pts

(d) 2 pts

Test continues on the other side of the sheet.

MATH 101 **25229**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. For this function, find f(-2), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 3\|x - 2\| - 3$	4. (a) 6 pts (b) 7 pts (c) 7 pts
5. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (3x + 2)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 2$ $B.\quad f(x) = 3x + 2,\ g(x) = x^2 + 4$ $C.\quad f(x) = x^2 + 2,\ g(x) = 3x + 4$ $D.\quad f(x) = 3x^2 + 4,\ g(x) = x + 2$ $(b)\quad F(x) = x^2,\ G(x) = x - 3,\ x_0 = 4$	5. (a) 8 pts (b) 8 pts (c) 4 pts

25229

TEST PAGE 1

1. (a)

$\frac{1}{8}$ 6 pts

(b)

$(-3, 0) \cup (3, 5)$ 4 pts

(c)

$(0, 3)$ 4 pts

(d)

$(0,\ 3)$ 3 pts

(e)

$(3,\ -3)$ 3 pts

2.

B

20 pts

3. (a)

$f(x) = 3x$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{3}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

TEST PAGE 2

4. (a)

9 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-3, \infty)$ 7 pts

5. (a)

A

8 pts

(b)

$x^2 - 6x + 9$

8 pts

(c)

1

4 pts

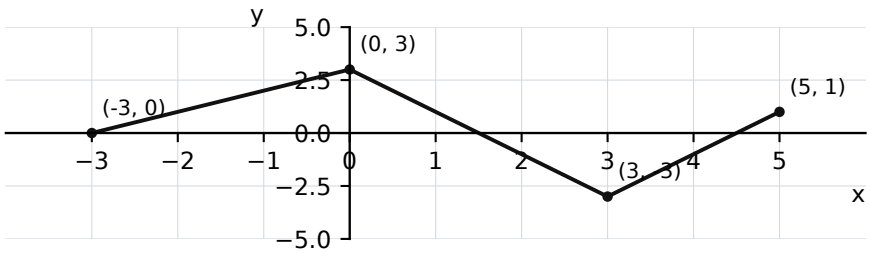
MATH 101 **36755**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-2), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 3\|x - 2\| - 3$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (3x + 2)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 2$ $B.\quad f(x) = 3x + 2,\ g(x) = x^2 + 4$ $C.\quad f(x) = x^2 + 2,\ g(x) = 3x + 4$ $D.\quad f(x) = 3x^2 + 4,\ g(x) = x + 2$ $(b)\quad F(x) = x^2,\ G(x) = x - 3,\ x_0 = 4$	2. (a) 8 pts (b) 8 pts (c) 4 pts
3. From the graph, find average rate from x = -3 to x = 5; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	3. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts

Test continues on the other side of the sheet.

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

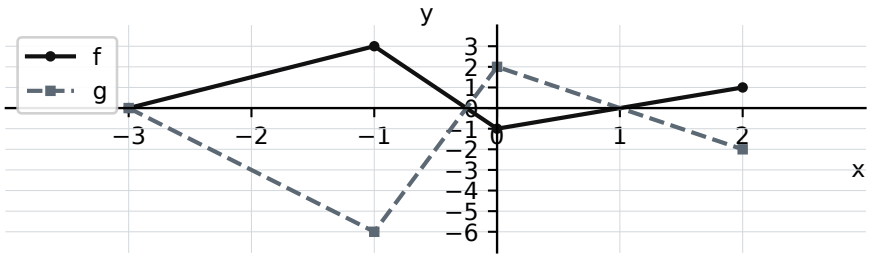
4. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x) - 2$

B. $g(x) = -2f(x)$

C. $g(x) = 2f(x)$

D. $g(x) = f(-2x)$



4.

20 pts

5. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 3);\ (3,\ 9)$

5. (a)

10 pts

(b)

6 pts

(c)

2 pts

(d)

2 pts

36755

TEST PAGE 1

1. (a)

9 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-3, \infty)$ — 7 pts

2. (a)

A — 8 pts

(b)

$x^2 - 6x + 9$ — 8 pts

(c)

1 — 4 pts

3. (a)

$\frac{1}{8}$ — 6 pts

(b)

$(-3, 0) \cup (3, 5)$ — 4 pts

(c)

$(0, 3)$ — 4 pts

(d)

$(0,\ 3)$ — 3 pts

(e)

$(3,\ -3)$ — 3 pts

TEST PAGE 2

4.

B — 20 pts

5. (a)

$f(x) = 3x$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{3}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

MATH 101 **81346**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

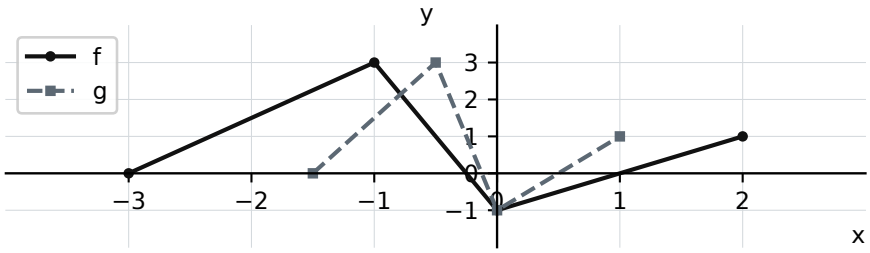
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(x-2)$

$B.\quad g(x) = f(2x)$

$C.\quad g(x) = f(x/2)$

$D.\quad g(x) = 2f(x)$



1.

20 pts

2. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (3x+2)^2 + 4$

$A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 2$

$B.\quad f(x) = 3x + 2,\ g(x) = x^2 + 4$

$C.\quad f(x) = x^2 + 2,\ g(x) = 3x + 4$

$D.\quad f(x) = 3x^2 + 4,\ g(x) = x + 2$

$(b)\quad F(x) = x^2,\ G(x) = x - 3,\ x_0 = 4$

2. (a)

8 pts

(b)

8 pts

(c)

4 pts

3. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 3);\ (3,\ 9)$

3. (a)

10 pts

(b)

6 pts

(c)

2 pts

(d)

2 pts

Test continues on the other side of the sheet.

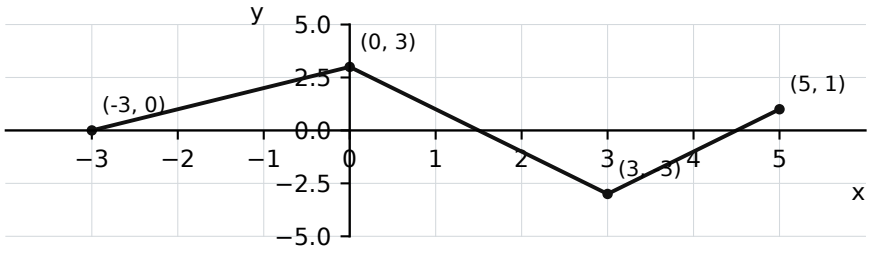
MATH 101 **81346**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -3 to x = 5; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	4. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts
5. For this function, find f(-2), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 3\|x - 2\| - 3$	5. (a) 6 pts (b) 7 pts (c) 7 pts

81346

TEST PAGE 1

1.

B

20 pts

2. (a)

A

8 pts

(b)

$x^2 - 6x + 9$

8 pts

(c)

1

4 pts

3. (a)

$f(x) = 3x$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{3}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

TEST PAGE 2

4. (a)

$\frac{1}{8}$ 6 pts

(b)

$(-3, 0) \cup (3, 5)$ 4 pts

(c)

$(0, 3)$ 4 pts

(d)

$(0,\ 3)$ 3 pts

(e)

$(3,\ -3)$ 3 pts

5. (a)

9 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

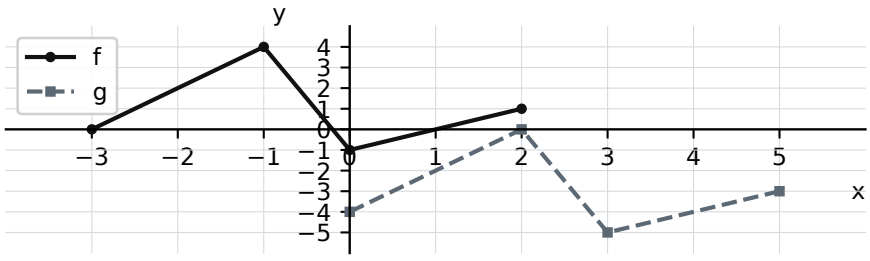
$[-3, \infty)$ 7 pts

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-3), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 4\|x - 3\| - 4$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x - 3) + 4$ $B.\quad g(x) = f(x + 3) - 4$ $C.\quad g(x) = f(x - 3) - 4$ $D.\quad g(x) = f(x + 3) + 4$ 	2. 20 pts
3. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 5);\ (3,\ 13)$	3. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts

Test continues on the other side of the sheet.

MATH 101 **91776**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (4x + 3)^2 + 5$

$A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 3$

$B.\quad f(x) = 4x + 3,\ g(x) = x^2 + 5$

$C.\quad f(x) = x^2 + 3,\ g(x) = 4x + 5$

$D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 3$

$(b)\quad F(x) = x^2,\ G(x) = x - 4,\ x_0 = 5$

4. (a)

8 pts

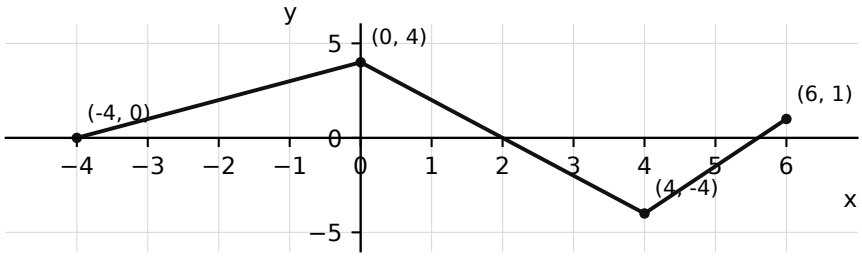
(b)

8 pts

(c)

4 pts

5. From the graph, find average rate from x = -4 to x = 6; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



5. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

91776

TEST PAGE 1

1. (a)

20 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-4, \infty)$ — 7 pts

2.

C

20 pts

3. (a)

$f(x) = 4x + 1$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{4} - \frac{1}{4}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

TEST PAGE 2

4. (a)

A

8 pts

(b)

$x^2 - 8x + 16$

8 pts

(c)

1

4 pts

5. (a)

$\frac{1}{10}$ — 6 pts

(b)

$(-4, 0) \cup (4, 6)$ — 4 pts

(c)

$(0, 4)$ — 4 pts

(d)

$(0,\ 4)$ — 3 pts

(e)

$(4,\ -4)$ — 3 pts

MATH 101 **52287**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-3), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 4\|x - 3\| - 4$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 5);\ (3,\ 13)$	2. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
3. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (4x + 3)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 3$ $B.\quad f(x) = 4x + 3,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 3,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 3$ $(b)\quad F(x) = x^2,\ G(x) = x - 4,\ x_0 = 5$	3. (a) 8 pts (b) 8 pts (c) 4 pts

Test continues on the other side of the sheet.

MATH 101 **52287**

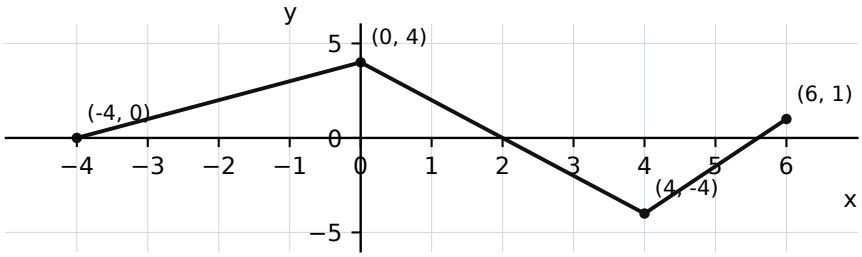
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -4 to x = 6; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



4. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

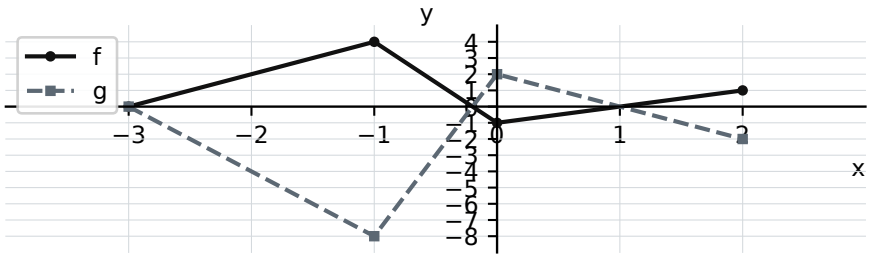
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = 2f(x)$

B. $g(x) = f(-2x)$

C. $g(x) = -2f(x)$

D. $g(x) = f(x) - 2$



5.

20 pts

52287

TEST PAGE 1

1. (a)

20 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-4, \infty)$ — 7 pts

2. (a)

$f(x) = 4x + 1$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{4} - \frac{1}{4}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

3. (a)

A

8 pts

(b)

$x^2 - 8x + 16$

8 pts

(c)

1

4 pts

TEST PAGE 2

4. (a)

$\frac{1}{10}$ — 6 pts

(b)

$(-4, 0) \cup (4, 6)$ — 4 pts

(c)

$(0, 4)$ — 4 pts

(d)

$(0,\ 4)$ — 3 pts

(e)

$(4,\ -4)$ — 3 pts

5.

C

20 pts

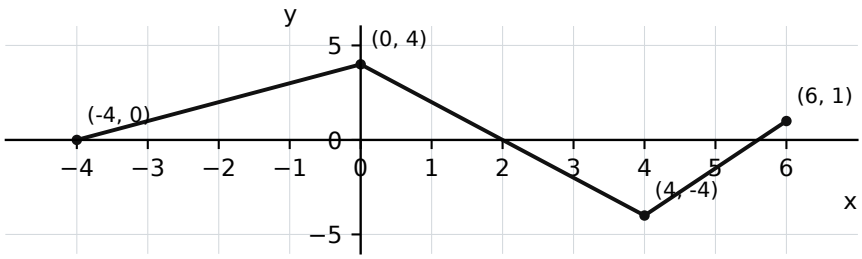
MATH 101 **48295**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. From the graph, find average rate from x = -4 to x = 6; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	1. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts
2. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (4x+3)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 3$ $B.\quad f(x) = 4x + 3,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 3,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 3$ $(b)\quad F(x) = x^2,\ G(x) = x - 4,\ x_0 = 5$	2. (a) 8 pts (b) 8 pts (c) 4 pts
3. For this function, find f(-3), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 4\lvert x - 3\rvert - 4$	3. (a) 6 pts (b) 7 pts (c) 7 pts

Test continues on the other side of the sheet.

MATH 101 **48295**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

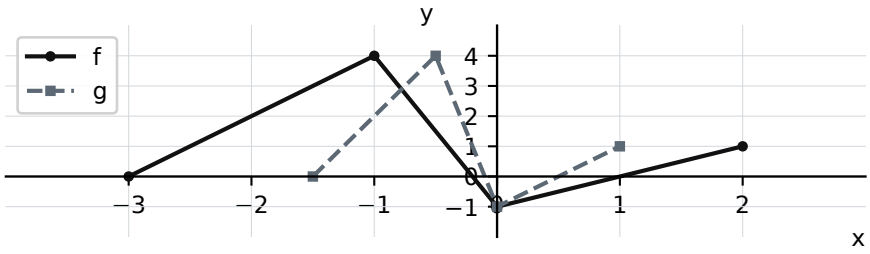
4. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x/2)$

B. $g(x) = 2f(x)$

C. $g(x) = f(2x)$

D. $g(x) = f(x - 2)$



4.

20 pts

5. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 5);\ (3,\ 13)$

5. (a)

10 pts

(b)

6 pts

(c)

2 pts

(d)

2 pts

48295

TEST PAGE 1

1. (a)

$\frac{1}{10}$ 6 pts

(b)

$(-4, 0) \cup (4, 6)$ 4 pts

(c)

$(0, 4)$ 4 pts

(d)

$(0,\ 4)$ 3 pts

(e)

$(4,\ -4)$ 3 pts

2. (a)

A

8 pts

(b)

$x^2 - 8x + 16$

8 pts

(c)

1

4 pts

3. (a)

20 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-4, \infty)$ 7 pts

TEST PAGE 2

4.

C

20 pts

5. (a)

$f(x) = 4x + 1$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{4} - \frac{1}{4}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

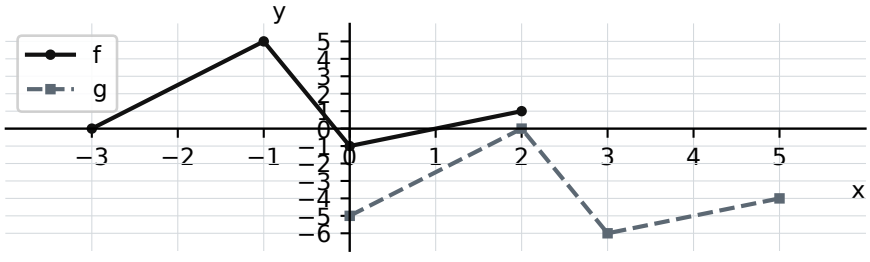
MATH 101 **75925**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 7);\ (3,\ 17)$	1. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
2. For this function, find f(-4), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 5\|x-4\| - 5$	2. (a) 6 pts (b) 7 pts (c) 7 pts
3. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x-3) - 5$ $B.\quad g(x) = f(x+3) + 5$ $C.\quad g(x) = f(x-3) + 5$ $D.\quad g(x) = f(x+3) - 5$ 	3. 20 pts

Test continues on the other side of the sheet.

MATH 101 **75925**

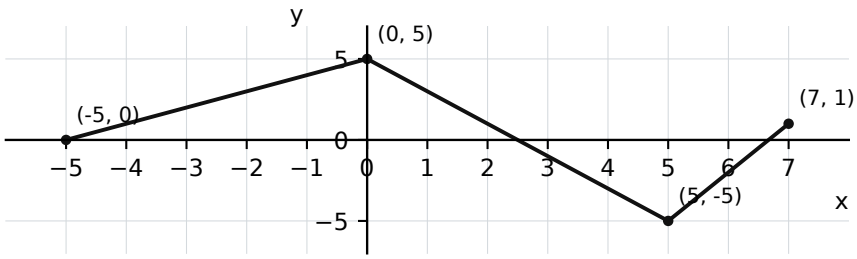
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -5 to x = 7; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



4. (a) 6 pts

(b) 4 pts

(c) 4 pts

(d) 3 pts

(e) 3 pts

5. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (5x + 4)^2 + 6$

$A.\quad f(x) = x^2 + 4,\ g(x) = 5x + 6$

$B.\quad f(x) = 5x^2 + 6,\ g(x) = x + 4$

$C.\quad f(x) = x^2 + 6,\ g(x) = 5x + 4$

$D.\quad f(x) = 5x + 4,\ g(x) = x^2 + 6$

$(b)\quad F(x) = x^2,\ G(x) = x - 5,\ x_0 = 6$

5. (a) 8 pts

(b) 8 pts

(c) 4 pts

75925

TEST PAGE 1

1. (a)

$f(x) = 5x + 2$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{5} - \frac{2}{5}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

2. (a)

35 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-5, \infty)$ — 7 pts

3.

A

20 pts

TEST PAGE 2

4. (a)

$\frac{1}{12}$ — 6 pts

(b)

$(-5, 0) \cup (5, 7)$ — 4 pts

(c)

$(0, 5)$ — 4 pts

(d)

$(0,\ 5)$ — 3 pts

(e)

$(5,\ -5)$ — 3 pts

5. (a)

C

8 pts

(b)

$x^2 - 10x + 25$

8 pts

(c)

1

4 pts

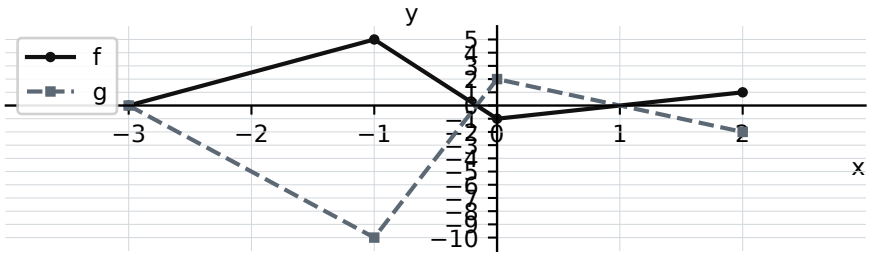
MATH 101 **75388**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-4), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 5\|x - 4\| - 5$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = -2f(x)$ $B.\quad g(x) = f(x) - 2$ $C.\quad g(x) = 2f(x)$ $D.\quad g(x) = f(-2x)$ 	2. 20 pts
3. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 7);\ (3,\ 17)$	3. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts

Test continues on the other side of the sheet.

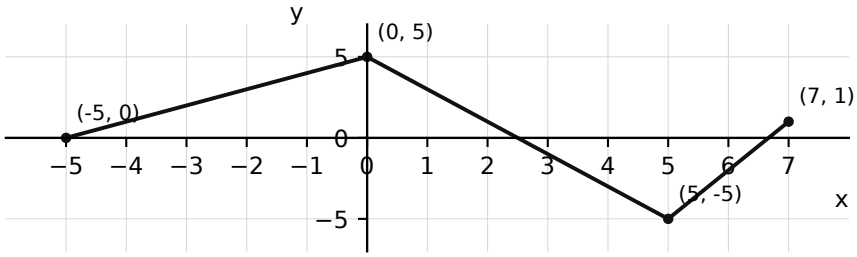
MATH 101 **75388**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (5x + 4)^2 + 6$ $A.\quad f(x) = x^2 + 4,\ g(x) = 5x + 6$ $B.\quad f(x) = 5x^2 + 6,\ g(x) = x + 4$ $C.\quad f(x) = x^2 + 6,\ g(x) = 5x + 4$ $D.\quad f(x) = 5x + 4,\ g(x) = x^2 + 6$ $(b)\quad F(x) = x^2,\ G(x) = x - 5,\ x_0 = 6$	4. (a) 8 pts (b) 8 pts (c) 4 pts
5. From the graph, find average rate from x = -5 to x = 7; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	5. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts

75388

TEST PAGE 1

1. (a)

35 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-5, \infty)$ — 7 pts

2.

A

20 pts

3. (a)

$f(x) = 5x + 2$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{5} - \frac{2}{5}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

TEST PAGE 2

4. (a)

C

8 pts

(b)

$x^2 - 10x + 25$

8 pts

(c)

1

4 pts

5. (a)

$\frac{1}{12}$ — 6 pts

(b)

$(-5, 0) \cup (5, 7)$ — 4 pts

(c)

$(0, 5)$ — 4 pts

(d)

$(0,\ 5)$ — 3 pts

(e)

$(5,\ -5)$ — 3 pts

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (5x + 4)^2 + 6$

A. $f(x) = x^2 + 4,\ g(x) = 5x + 6$

B. $f(x) = 5x^2 + 6,\ g(x) = x + 4$

C. $f(x) = x^2 + 6,\ g(x) = 5x + 4$

D. $f(x) = 5x + 4,\ g(x) = x^2 + 6$

(*b*) $F(x) = x^2,\ G(x) = x - 5,\ x_0 = 6$

1. (a)

8 pts

(b)

8 pts

(c)

4 pts

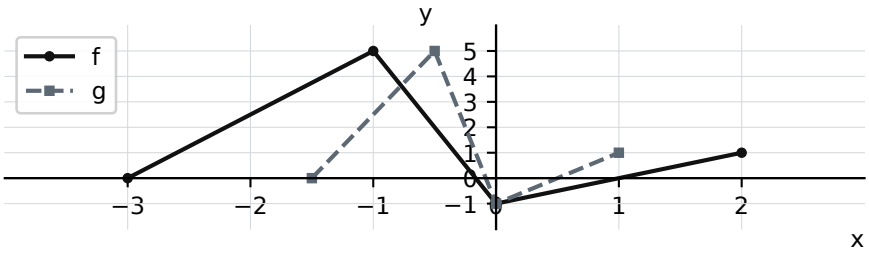
2. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(2x)$

B. $g(x) = f(x - 2)$

C. $g(x) = f(x/2)$

D. $g(x) = 2f(x)$



2.

20 pts

3. For this function, find f(-4), domain, and range. Record: (a) value; (b) domain; (c) range.

$f(x) = 5|x - 4| - 5$

3. (a)

6 pts

(b)

7 pts

(c)

7 pts

Test continues on the other side of the sheet.

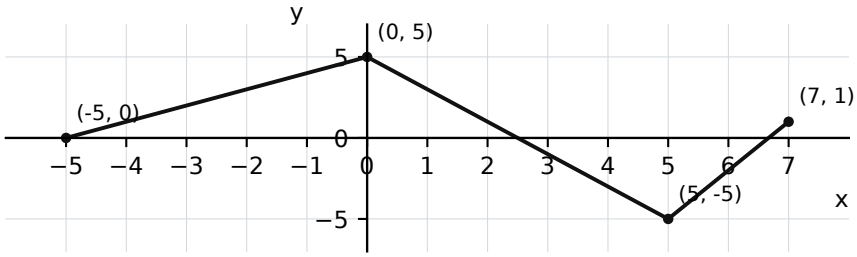
MATH 101 **40170**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -5 to x = 7; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	4. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts
5. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 7);\ (3,\ 17)$	5. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts

40170

TEST PAGE 1

1. (a)

C

8 pts

(b)

$x^2 - 10x + 25$

8 pts

(c)

1

4 pts

2.

A

20 pts

3. (a)

35

6 pts

(b)

$(-\infty, \infty)$

7 pts

(c)

$[-5, \infty)$

7 pts

TEST PAGE 2

4. (a)

$\frac{1}{12}$

6 pts

(b)

$(-5, 0) \cup (5, 7)$

4 pts

(c)

$(0, 5)$

4 pts

(d)

$(0,\ 5)$

3 pts

(e)

$(5,\ -5)$

3 pts

5. (a)

$f(x) = 5x + 2$

10 pts

(b)

$f^{-1}(x) = \frac{x}{5} - \frac{2}{5}$

6 pts

(c)

$(-\infty, \infty)$

2 pts

(d)

$(-\infty, \infty)$

2 pts

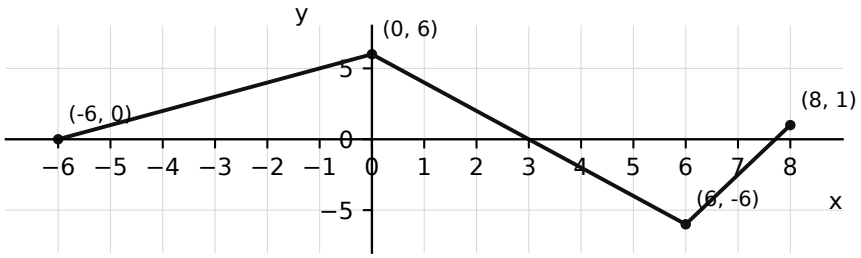
MATH 101 **23259**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-5), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 6\|x - 5\| - 6$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 9);\ (3,\ 21)$	2. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
3. From the graph, find average rate from x = -6 to x = 8; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	3. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts

Test continues on the other side of the sheet.

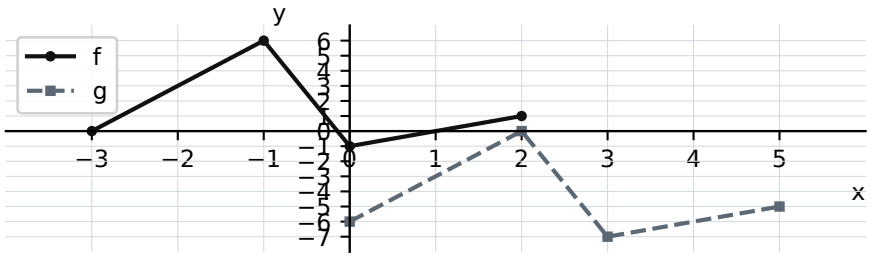
MATH 101 **23259**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (6x + 5)^2 + 7$ $A.\quad f(x) = x^2 + 5,\ g(x) = 6x + 7$ $B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 5$ $C.\quad f(x) = 6x^2 + 7,\ g(x) = x + 5$ $D.\quad f(x) = 6x + 5,\ g(x) = x^2 + 7$ $(b)\quad F(x) = x^2,\ G(x) = x - 6,\ x_0 = 7$	4. (a) 8 pts (b) 8 pts (c) 4 pts
5. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x - 3) + 6$ $B.\quad g(x) = f(x + 3) - 6$ $C.\quad g(x) = f(x + 3) + 6$ $D.\quad g(x) = f(x - 3) - 6$ 	5. 20 pts

23259

TEST PAGE 1

1. (a)

54 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-6, \infty)$ — 7 pts

2. (a)

$f(x) = 6x + 3$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{6} - \frac{1}{2}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

3. (a)

$\frac{1}{14}$ — 6 pts

(b)

$(-6, 0) \cup (6, 8)$ — 4 pts

(c)

$(0, 6)$ — 4 pts

(d)

$(0,\ 6)$ — 3 pts

(e)

$(6,\ -6)$ — 3 pts

TEST PAGE 2

4. (a)

B — 8 pts

(b)

$x^2 - 12x + 36$ — 8 pts

(c)

1 — 4 pts

5.

D — 20 pts

MATH 101 **47362**

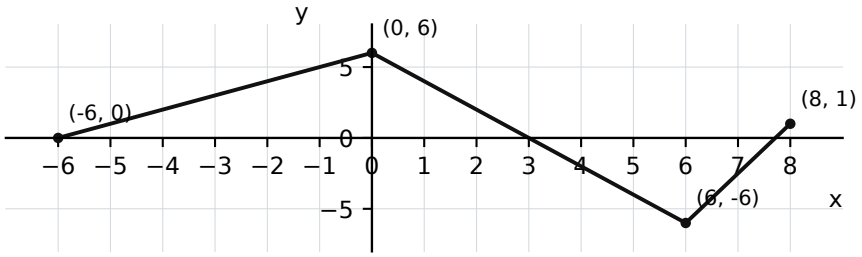
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. From the graph, find average rate from x = -6 to x = 8; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



1. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

2. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value.

$h(x) = (6x+5)^2 + 7$

$A.\quad f(x) = x^2 + 5,\ g(x) = 6x + 7$

$B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 5$

$C.\quad f(x) = 6x^2 + 7,\ g(x) = x + 5$

$D.\quad f(x) = 6x + 5,\ g(x) = x^2 + 7$

$(b)\quad F(x) = x^2,\ G(x) = x - 6,\ x_0 = 7$

2. (a)

8 pts

(b)

8 pts

(c)

4 pts

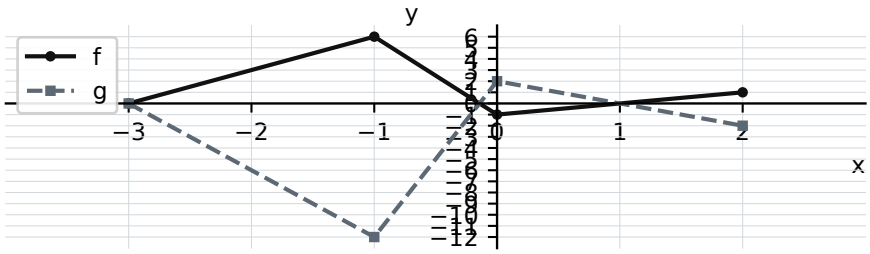
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = 2f(x)$

$B.\quad g(x) = f(-2x)$

$C.\quad g(x) = f(x) - 2$

$D.\quad g(x) = -2f(x)$



3.

20 pts

Test continues on the other side of the sheet.

MATH 101 **47362**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 9);\ (3,\ 21)$	4. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
5. For this function, find f(-5), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 6\|x - 5\| - 6$	5. (a) 6 pts (b) 7 pts (c) 7 pts

47362

TEST PAGE 1

1. (a)

$\frac{1}{14}$ 6 pts

(b)

$(-6, 0) \cup (6, 8)$ 4 pts

(c)

$(0, 6)$ 4 pts

(d)

$(0,\ 6)$ 3 pts

(e)

$(6,\ -6)$ 3 pts

2. (a)

B

8 pts

(b)

$x^2 - 12x + 36$

8 pts

(c)

1

4 pts

3.

D

20 pts

TEST PAGE 2

4. (a)

$f(x) = 6x + 3$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{6} - \frac{1}{2}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

5. (a)

54 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-6, \infty)$ 7 pts

MATH 101 **41802**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

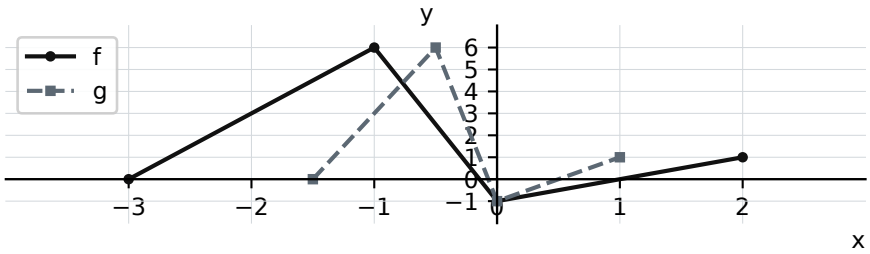
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x/2)$

B. $g(x) = 2f(x)$

C. $g(x) = f(x - 2)$

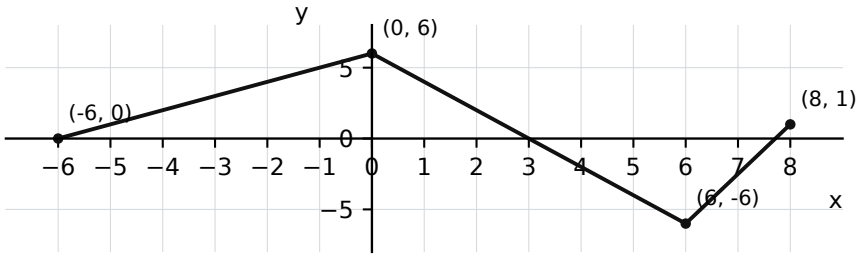
D. $g(x) = f(2x)$



1.

20 pts

2. From the graph, find average rate from x = -6 to x = 8; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



2. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

3. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 9);\ (3,\ 21)$

3. (a)

10 pts

(b)

6 pts

(c)

2 pts

(d)

2 pts

Test continues on the other side of the sheet.

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (6x + 5)^2 + 7$ $A.\quad f(x) = x^2 + 5,\ g(x) = 6x + 7$ $B.\quad f(x) = x^2 + 7,\ g(x) = 6x + 5$ $C.\quad f(x) = 6x^2 + 7,\ g(x) = x + 5$ $D.\quad f(x) = 6x + 5,\ g(x) = x^2 + 7$ $(b)\quad F(x) = x^2,\ G(x) = x - 6,\ x_0 = 7$	4. (a) 8 pts (b) 8 pts (c) 4 pts
5. For this function, find f(-5), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 6\lvert x - 5\rvert - 6$	5. (a) 6 pts (b) 7 pts (c) 7 pts

41802

TEST PAGE 1

1.

D

20 pts

2. (a)

$\frac{1}{14}$ 6 pts

(b)

$(-6, 0) \cup (6, 8)$ 4 pts

(c)

$(0, 6)$ 4 pts

(d)

$(0,\ 6)$ 3 pts

(e)

$(6,\ -6)$ 3 pts

3. (a)

$f(x) = 6x + 3$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{6} - \frac{1}{2}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

TEST PAGE 2

4. (a)

B

8 pts

(b)

$x^2 - 12x + 36$

8 pts

(c)

1

4 pts

5. (a)

54 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-6, \infty)$ 7 pts

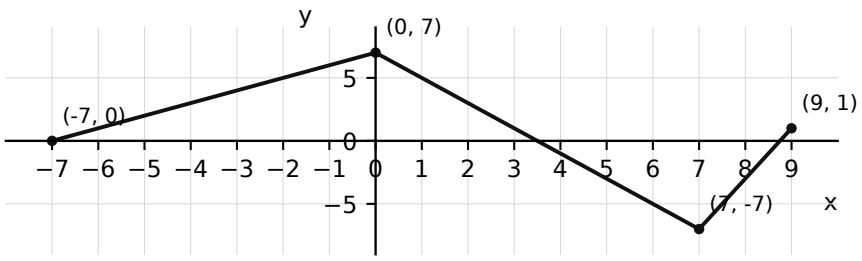
MATH 101 **24714**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-6), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 7\|x - 6\| - 7$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (7x + 6)^2 + 8$ $A.\quad f(x) = x^2 + 8,\ g(x) = 7x + 6$ $B.\quad f(x) = 7x^2 + 8,\ g(x) = x + 6$ $C.\quad f(x) = 7x + 6,\ g(x) = x^2 + 8$ $D.\quad f(x) = x^2 + 6,\ g(x) = 7x + 8$ $(b)\quad F(x) = x^2,\ G(x) = x - 7,\ x_0 = 8$	2. (a) 8 pts (b) 8 pts (c) 4 pts
3. From the graph, find average rate from x = -7 to x = 9; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min. 	3. (a) 6 pts (b) 4 pts (c) 4 pts (d) 3 pts (e) 3 pts

Test continues on the other side of the sheet.

MATH 101 **24714**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

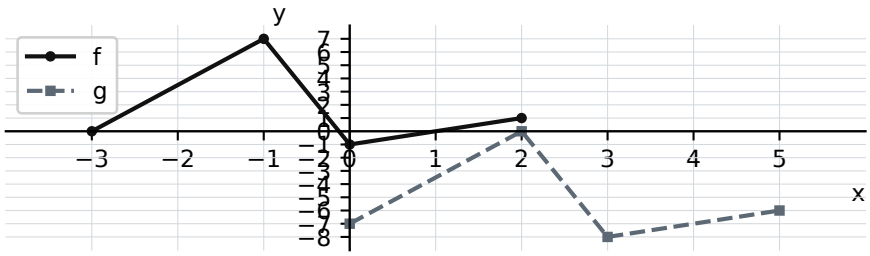
4. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-3) - 7$

B. $g(x) = f(x+3) + 7$

C. $g(x) = f(x-3) + 7$

D. $g(x) = f(x+3) - 7$



4.

20 pts

5. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 11);\ (3,\ 25)$

5. (a)

10 pts

(b)

6 pts

(c)

2 pts

(d)

2 pts

24714

TEST PAGE 1

1. (a)

77 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-7, \infty)$ — 7 pts

2. (a)

A

8 pts

(b)

$x^2 - 14x + 49$

8 pts

(c)

1

4 pts

3. (a)

$\frac{1}{16}$ — 6 pts

(b)

$(-7, 0) \cup (7, 9)$ — 4 pts

(c)

$(0, 7)$ — 4 pts

(d)

$(0,\ 7)$ — 3 pts

(e)

$(7,\ -7)$ — 3 pts

TEST PAGE 2

4.

A

20 pts

5. (a)

$f(x) = 7x + 4$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{7} - \frac{4}{7}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

MATH 101 **36777**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

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1. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range.

$(1,\ 11);\ (3,\ 25)$

1. (a)

10 pts

(b)

6 pts

(c)

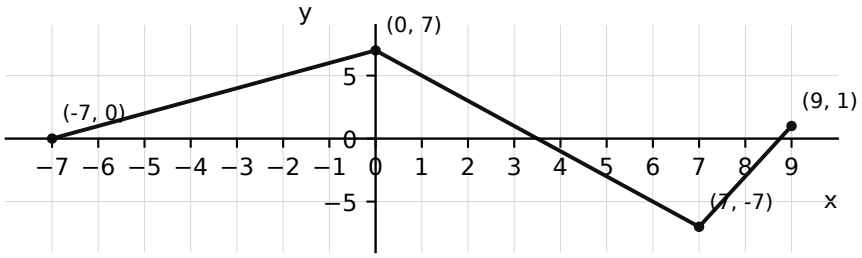
2 pts

(d)

2 pts

---

2. From the graph, find average rate from x = -7 to x = 9; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



2. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

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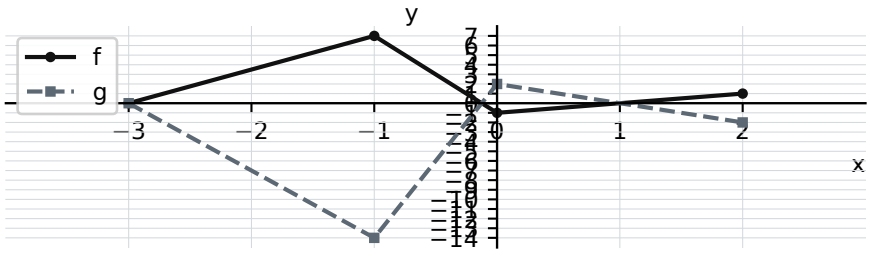
3. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = -2f(x)$

$B.\quad g(x) = f(x) - 2$

$C.\quad g(x) = 2f(x)$

$D.\quad g(x) = f(-2x)$



3.

20 pts

Test continues on the other side of the sheet.

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (7x + 6)^2 + 8$ $A.\quad f(x) = x^2 + 8,\ g(x) = 7x + 6$ $B.\quad f(x) = 7x^2 + 8,\ g(x) = x + 6$ $C.\quad f(x) = 7x + 6,\ g(x) = x^2 + 8$ $D.\quad f(x) = x^2 + 6,\ g(x) = 7x + 8$ $(b)\quad F(x) = x^2,\ G(x) = x - 7,\ x_0 = 8$	4. (a) 8 pts (b) 8 pts (c) 4 pts
5. For this function, find f(-6), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 7\lvert x - 6\rvert - 7$	5. (a) 6 pts (b) 7 pts (c) 7 pts

36777

TEST PAGE 1

1. (a)

$f(x) = 7x + 4$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{7} - \frac{4}{7}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

2. (a)

$\frac{1}{16}$ — 6 pts

(b)

$(-7, 0) \cup (7, 9)$ — 4 pts

(c)

$(0, 7)$ — 4 pts

(d)

$(0,\ 7)$ — 3 pts

(e)

$(7,\ -7)$ — 3 pts

3.

A

20 pts

TEST PAGE 2

4. (a)

A

8 pts

(b)

$x^2 - 14x + 49$

8 pts

(c)

1

4 pts

5. (a)

77 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-7, \infty)$ — 7 pts

MATH 101 **68899**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. For this function, find f(-6), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 7\|x - 6\| - 7$	1. (a) 6 pts (b) 7 pts (c) 7 pts
2. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 11);\ (3,\ 25)$	2. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
3. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (7x + 6)^2 + 8$ $A.\quad f(x) = x^2 + 8,\ g(x) = 7x + 6$ $B.\quad f(x) = 7x^2 + 8,\ g(x) = x + 6$ $C.\quad f(x) = 7x + 6,\ g(x) = x^2 + 8$ $D.\quad f(x) = x^2 + 6,\ g(x) = 7x + 8$ $(b)\quad F(x) = x^2,\ G(x) = x - 7,\ x_0 = 8$	3. (a) 8 pts (b) 8 pts (c) 4 pts

Test continues on the other side of the sheet.

MATH 101 **68899**

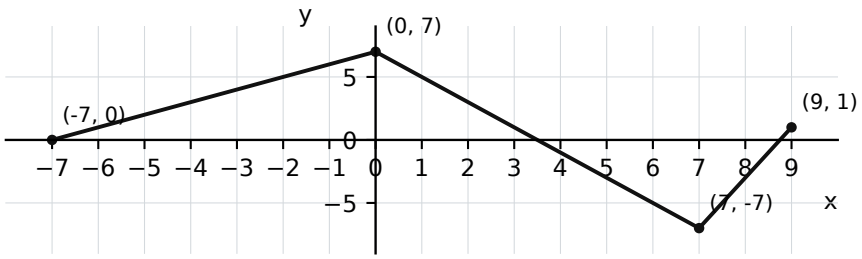
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -7 to x = 9; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



4. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

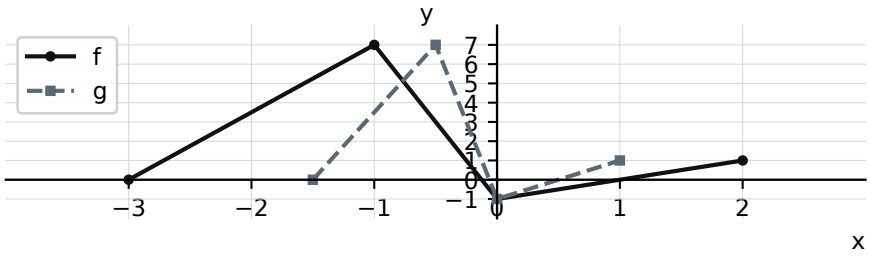
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

$A.\quad g(x) = f(2x)$

$B.\quad g(x) = f(x-2)$

$C.\quad g(x) = f(x/2)$

$D.\quad g(x) = 2f(x)$



5.

20 pts

68899

TEST PAGE 1

1. (a)

77 — 6 pts

(b)

$(-\infty, \infty)$ — 7 pts

(c)

$[-7, \infty)$ — 7 pts

2. (a)

$f(x) = 7x + 4$ — 10 pts

(b)

$f^{-1}(x) = \frac{x}{7} - \frac{4}{7}$ — 6 pts

(c)

$(-\infty, \infty)$ — 2 pts

(d)

$(-\infty, \infty)$ — 2 pts

3. (a)

A

8 pts

(b)

$x^2 - 14x + 49$

8 pts

(c)

1

4 pts

TEST PAGE 2

4. (a)

$\frac{1}{16}$ — 6 pts

(b)

$(-7, 0) \cup (7, 9)$ — 4 pts

(c)

$(0, 7)$ — 4 pts

(d)

$(0,\ 7)$ — 3 pts

(e)

$(7,\ -7)$ — 3 pts

5.

A

20 pts

MATH 101 **58437**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

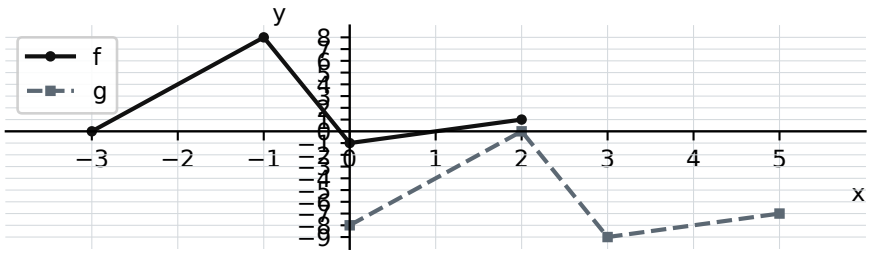
1. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x+3) + 8$

B. $g(x) = f(x-3) + 8$

C. $g(x) = f(x+3) - 8$

D. $g(x) = f(x-3) - 8$



1.

20 pts

2. For this function, find f(-7), domain, and range. Record: (a) value; (b) domain; (c) range.

$f(x) = 8|x-7| - 8$

2. (a)

6 pts

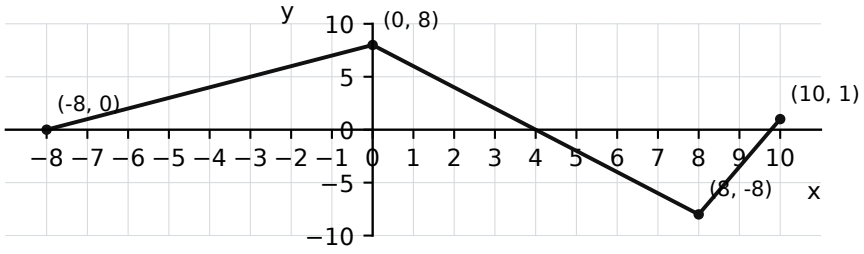
(b)

7 pts

(c)

7 pts

3. From the graph, find average rate from x = -8 to x = 10; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



3. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

Test continues on the other side of the sheet.

MATH 101 **58437**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 13);\ (3,\ 29)$	4. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
5. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (8x + 7)^2 + 9$ $A.\quad f(x) = x^2 + 9,\ g(x) = 8x + 7$ $B.\quad f(x) = x^2 + 7,\ g(x) = 8x + 9$ $C.\quad f(x) = 8x^2 + 9,\ g(x) = x + 7$ $D.\quad f(x) = 8x + 7,\ g(x) = x^2 + 9$ $(b)\quad F(x) = x^2,\ G(x) = x - 8,\ x_0 = 9$	5. (a) 8 pts (b) 8 pts (c) 4 pts

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TEST PAGE 1

1.

D

20 pts

2. (a)

104 6 pts

(b)

$(-\infty, \infty)$ 7 pts

(c)

$[-8, \infty)$ 7 pts

3. (a)

$\frac{1}{18}$ 6 pts

(b)

$(-8, 0) \cup (8, 10)$ 4 pts

(c)

$(0, 8)$ 4 pts

(d)

$(0,\ 8)$ 3 pts

(e)

$(8,\ -8)$ 3 pts

TEST PAGE 2

4. (a)

$f(x) = 8x + 5$ 10 pts

(b)

$f^{-1}(x) = \frac{x}{8} - \frac{5}{8}$ 6 pts

(c)

$(-\infty, \infty)$ 2 pts

(d)

$(-\infty, \infty)$ 2 pts

5. (a)

A

8 pts

(b)

$x^2 - 16x + 64$

8 pts

(c)

1

4 pts

MATH 101 **71830**

Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Choose the pair giving h= f(g). Form F(G) and evaluate it at the supplied input. Record: (a) choice; (b) composition; (c) value. $h(x) = (8x + 7)^2 + 9$ $A.\quad f(x) = x^2 + 9,\ g(x) = 8x + 7$ $B.\quad f(x) = x^2 + 7,\ g(x) = 8x + 9$ $C.\quad f(x) = 8x^2 + 9,\ g(x) = x + 7$ $D.\quad f(x) = 8x + 7,\ g(x) = x^2 + 9$ $(b)\quad F(x) = x^2,\ G(x) = x - 8,\ x_0 = 9$	1. (a) 8 pts (b) 8 pts (c) 4 pts
2. Find the linear function through these points, with domain all real numbers. Find its inverse and give the inverse domain and range. Record: (a) function; (b) inverse; (c) domain; (d) range. $(1,\ 13);\ (3,\ 29)$	2. (a) 10 pts (b) 6 pts (c) 2 pts (d) 2 pts
3. For this function, find f(-7), domain, and range. Record: (a) value; (b) domain; (c) range. $f(x) = 8\lvert x - 7\rvert - 8$	3. (a) 6 pts (b) 7 pts (c) 7 pts

Test continues on the other side of the sheet.

MATH 101 **71830**

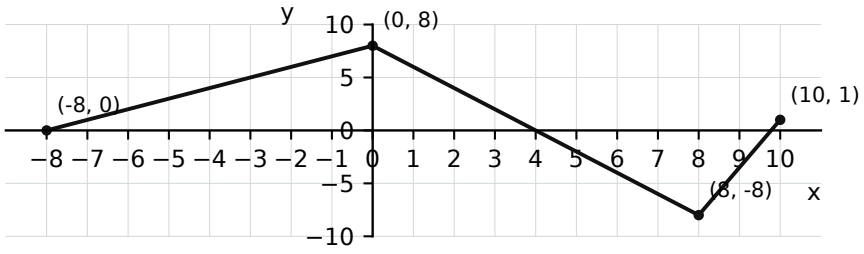
Unit 2: Functions and Their Graphs

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

4. From the graph, find average rate from x = -8 to x = 10; increasing and decreasing intervals; the two local extrema (ordered pairs). Record: (a) rate; (b) increasing; (c) decreasing; (d) local max; (e) local min.



4. (a)

6 pts

(b)

4 pts

(c)

4 pts

(d)

3 pts

(e)

3 pts

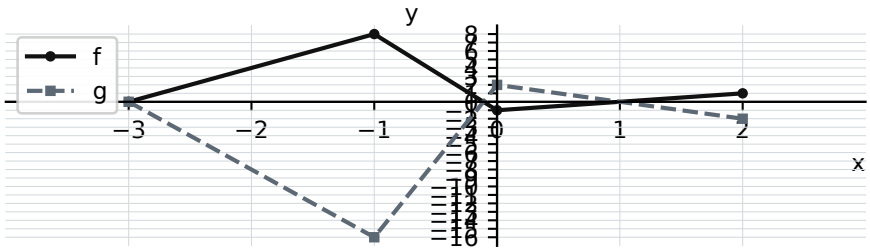
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x) - 2$

B. $g(x) = 2f(x)$

C. $g(x) = f(-2x)$

D. $g(x) = -2f(x)$



5.

20 pts

71830

TEST PAGE 1

1. (a)

A

8 pts

(b)

$x^2 - 16x + 64$

8 pts

(c)

1

4 pts

2. (a)

$f(x) = 8x + 5$

10 pts

(b)

$f^{-1}(x) = \frac{x}{8} - \frac{5}{8}$

6 pts

(c)

$(-\infty, \infty)$

2 pts

(d)

$(-\infty, \infty)$

2 pts

3. (a)

104

6 pts

(b)

$(-\infty, \infty)$

7 pts

(c)

$[-8, \infty)$

7 pts

TEST PAGE 2

4. (a)

$\frac{1}{18}$

6 pts

(b)

$(-8, 0) \cup (8, 10)$

4 pts

(c)

$(0, 8)$

4 pts

(d)

$(0,\ 8)$

3 pts

(e)

$(8,\ -8)$

3 pts

5.

D

20 pts